Before class: the organic acids and bases

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. The answer key is at the back.

1 · Read first (5 min)

Two parents

A **carboxylic acid** has a –COOH group: a C=O and an –OH on the same carbon. Vinegar is acetic acid. An **amine** is ammonia (NH_3) with carbon groups in place of H atoms. Acids give away H^+; amines pick it up. They are the acids and bases of organic chemistry.

Two children

An acid can join to an **alcohol**, losing water, to make an **ester** (–COO–). It can join to an **amine**, also losing water, to make an **amide** (–CONH–). Esters smell of fruit; amides hold proteins together. Water can split both apart again: **hydrolysis**.

Why a pharmacist cares

Aspirin is an ester and an acid. Paracetamol is an amide. Most medicines that act on the brain and heart are amines, sold as **hydrochloride** salts so they dissolve in water. Knowing these four groups tells you how a drug dissolves, how it breaks down on the shelf, and how the body handles it.

2 · Readiness check

Ten quick questions: recap and a preview.

Q1 Recap

Oxidation of a primary alcohol with excess acidified dichromate gives:

A. a ketone
B. a carboxylic acid
C. an ether
D. an alkene

Q2 Recap

Ammonia, NH_3, acts as a base because it:

A. has a lone pair that accepts H^+
B. gives away H^+
C. contains oxygen
D. is a gas

Q3 Recap

Which can form hydrogen bonds between its own molecules?

A. CH_3CH_2OH
B. CH_3OCH_3
C. $CH_3CH_2CH_3$
D. CH_3COCH_3

Q4 CH4-LO1

Match each group to its family.

Options: carboxylic acid · ester · amine · amide · ketone

–COOH _____

–COOR′ _____

–NH_2 _____

–$CONH_2$ _____

Q5 CH4-LO1

What is the common name of CH_3COOH, the acid in vinegar?

_____

Q6 CH4-LO2

Carboxylic acids are strong acids that ionise completely in water.

True / False

Q7 CH4-LO3

Which family gives bananas and pineapples their smell?

A. esters
B. amines
C. carboxylic acids
D. amides

Q8 CH4-LO4

Which smell comes from amines?

A. rotting fish
B. vinegar
C. bananas
D. nail-polish remover

Q9 CH4-LO5

Which drug contains an amide group?

A. paracetamol
B. ethanol
C. glycerol
D. acetone

Q10 CH4-LO3

An old bottle of aspirin tablets smells of vinegar. Suggest what has happened.

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PHD125 Pharmaceutical Organic Chemistry · Carboxylic Acid, Ester, Amines, Amide

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About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Answer key

2 · Readiness check

Q1 Recap

Oxidation of a primary alcohol with excess acidified dichromate gives:

A. a ketone

B. a carboxylic acid

C. an ether

D. an alkene

1° alcohol → aldehyde → carboxylic acid (Chapter 3).

Q2 Recap

Ammonia, NH_3, acts as a base because it:

A. has a lone pair that accepts H^+

B. gives away H^+

C. contains oxygen

D. is a gas

The lone pair on N accepts a proton to form NH_4^+.

Q3 Recap

Which can form hydrogen bonds between its own molecules?

A. CH_3CH_2OH

B. CH_3OCH_3

C. $CH_3CH_2CH_3$

D. CH_3COCH_3

It needs an O–H (or N–H) group.

Q4 CH4-LO1

Match each group to its family.

–COOH → carboxylic acid; –COOR′ → ester; $–NH_2$ → amine; $–CONH_2$ → amide

Ketone (R–CO–R′) is the extra option.

Q5 CH4-LO1

What is the common name of CH_3COOH, the acid in vinegar?

acetic acid

Acetic acid (IUPAC: ethanoic acid).

Q6 CH4-LO2

Carboxylic acids are strong acids that ionise completely in water.

True / False → **False**

They are weak acids: only a small fraction ionises.

Q7 CH4-LO3

Which family gives bananas and pineapples their smell?

A. esters

B. amines

C. carboxylic acids

D. amides

Isopentyl acetate (banana) and ethyl butyrate (pineapple).

Q8 CH4-LO4

Which smell comes from amines?

A. rotting fish

B. vinegar

C. bananas

D. nail-polish remover

Trimethylamine, putrescine and cadaverine.

Q9 CH4-LO5

Which drug contains an amide group?

A. paracetamol

B. ethanol

C. glycerol

D. acetone

Paracetamol is N-(4-hydroxyphenyl)acetamide.

Q10 CH4-LO3

An old bottle of aspirin tablets smells of vinegar. Suggest what has happened.

Moisture has slowly split (hydrolysed) the ester group in aspirin, giving salicylic acid and acetic acid. Acetic acid is the vinegar smell. The tablets have degraded and should be discarded (4.2).

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